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CLINICAL FEATURES AND TESTS FOR MALABSORPTION

Steps for evaluating malabsorption :

1. Confirm the diagnosis → Clinically / By Investigations
2. Confirm global vs selective malabsorption.
3. Confirm the etiology.

Clinical features

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System	Scenarios	Features
Gastrointestinal	Osmotic diarrhoea	Site : Small intestine Duration : >4 weeks Stool osmotic gap : ↑ Improves on fasting most consistent symptom
	Fat malabsorption	Steatorrhea
	Carbohydrate malabsorption	Abdominal bloating and dyspeptic symptoms
	Protein malabsorption	Foul smelling stools, ascites, edema
	Unexplained weight loss	-
Musculoskeletal	Fat malabsorption (D/t impaired vitamin D and Ca^{2+} absorption)	Tetany, fracture, bone pain
Cutaneous manifestations	Vitamin B_{12}, Niacin deficiency	Hyperpigmentation
	Vitamin C deficiency	Perifollicular hemorrhage, curly hair, spiral hair
	Vitamin K deficiency	Bruisability, ecchymoses
	Zn & Essential fatty acid deficiency	Acrodermatitis
	Vitamin A deficiency	Follicular hyperkeratosis
Miscellaneous	Coeliac disease	Growth retardation
	Anaemia (D/t intestinal malabsorption) : Distal pathology → Vitamin B_{12} deficiency Proximal pathology → Folate/Iron deficiency	-
	Vitamin B12/E deficiency Vitamin B12 deficiency Vitamin A deficiency	Peripheral neuropathy, ataxia Glossitis, cheliosis, stomatitis Night blindness, xerophthalmia

System	Scenarios	Features	
Miscellaneous	Kidney stones	Normal physiology : Oxalate + Ca^{2+} → Excreted	malabsorption : Fatty acid + Ca^{2+} → Free oxalate in gut → Oxalate absorbed in colon → Oxaluria → Ca^{2+} oxalate stones

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Note :

- Symptoms not seen in malabsorption : Abdominal pain, constipation, fever.
- Differentials of malabsorption (Similar symptoms) :
  - TB : Weight loss, long standing diarrhoea, edema/ascites.
  - Nephrotic syndrome : Protein loss.

Investigations

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FAT MALABSORPTION

- Qualitative test : Sudan III.
- Quantitative test (Gold standard) : Fecal fat excretion ≥7 g/day.

Note : Fecal fat excretion test is the gold standard Investigation for malabsorption.

CARBOHYDRATE MALABSORPTION

1. D-Xylose test :

- Best and most specific test.
- Composition of D-xylose : Pentose monosaccharide.
- Used to assess intact small intestinal mucosa (D-xylose absorbed here).
- Procedure :

25g of D-xylose given orally after fasting (Early morning)

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Collect urine sample (Entire volume passed) after 4-6 hours & check D-xylose levels

- >4.5 g → Interpretation : Normal
- <4.5 g (Positive) → Interpretation : Small intestinal mucosal pathology

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- False positives seen in (Etiology is not in small intestine) :

Clinical scenario	Reason for false positive
Renal failure	Defective excretion
Rapid intestinal transit (Hyperthyroidism, Vomiting)	Reduced time for absorption in small intestine
SIBO (Small Intestinal Bacterial Overgrowth)	Organisms utilise D-xylose → Reduced absorption
Ascites	Sequestration of D-xylose into the third space → Reduced absorption

2. Lactose tolerance test : Obsolete.

3. Lactose H_2 breath test :

- Done for lactase deficiency.
- Deficiency of lactase → Accumulation of lactose in large intestine → Bacterial action on lactose → H_2 production.
- Diagnosis : ↑ in breath H_2 >20 ppm.

4. Lactulose H_2 breath test :

- metabolised in large intestine to produce H_2.
- H_2 produced :
  - Within 90 minutes (Early peaking) : SIBO.
  - >90 minutes : Normal physiology.

OBSOLETE TESTS FOR MALABSORPTION

Test	Etiology
Acid steatocrit test	Fat malabsorption
C-14 Triolein breath test	
C-13 Egg white breath test (Phenol and p-cresol in urine)	Protein malabsorption

Note :

- Stool elastase, chymotrypsin : Pancreatic insufficiency.
- Fecal α-1 antitrypsin : Primary intestinal lymphangiectasia.
- NIRA (Near infrared reflectance analysis) : Newer test for malabsorption.
- Diseases diagnosed on biopsy :
  1. Whipple's disease.
  2. Agammaglobinemia.
  3. Abetalipoproteinemia.

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Schilling test :

Steps :

Phase I : 1000 µg of unlabelled Vitamin B_{12} I.M to saturate stores

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Give 1 µg of oral radiolabelled Vitamin B_{12}

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Check Vitamin B_{12} level in urine

- <7% → Positive test (malabsorption etiology)
- 7–10% → Normal

Phase II : Administer

- Intrinsic factor → If >10% urine B_{12} → Pernicious anemia (IF)
- Antibiotics → If >10% → SIBO
- Pancreatic enzymes → If >10% → Chronic pancreatitis

After phase I & 2, if urine B_{12} <7% = Ileal disease.